DAA ASSIGNMENT

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1)Write a program to find the sum of first n natural numbers using user-defined function.

```c
#include<stdio.h>
int sum_of_first_n_natural_numbers(int n){
int sum =0;
for(int i =1;i <=n ;i++){
sum = sum +i;
}
return sum;
}
int main(){
int n ;
printf("enter a value for n :");
scanf("%d",&n);
printf("The sum of first %d natural numbers %d ",n,sum_of_first_n_natural_numbers(n));
printf("\n");
return 0;
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc sum_sqr_num.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter a value for n :34
The sum of first square of 34 natural numbers 13685
```

Space Complexity:

- The loop uses only two variables (s and i), and no extra memory grows with n.
- So, the memory used stays constant no matter the input size.
- Therefore, the space complexity is O(1).

2)Write a program to find the sum of squares of first n natural numbers using user-defined function.

```c
#include<stdio.h>
int sum_of_square_of_first_n_natural_numbers(int n){
int sum =0;
for(int i =1;i <=n ;i++){
sum = sum +i*i;
}
return sum;
}
int main(){
int n ;
printf("enter a value for n :");
scanf("%d",&n);
printf("The sum of first square of %d natural numbers %d ",n,sum_of_square_of_first_n_natural_numbers(n));
printf("\n");
return 0;
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc sum_sqr_num.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter a value for n :34
The sum of first square of 34 natural numbers 13685
```

Space Complexity:

- The program uses only a few variables, and this number does not increase when n becomes bigger.
- Since the memory used stays the same all the time, the space complexity is O(1).

3)Write a program to find the sum of cubes of first n natural numbers.

```c
#include<stdio.h>
int sum_of_cubes_of_first_n_natural_numbers(int n){
int sum =0;
for(int i =1;i <=n ;i++){
sum = sum +i*i*i;
}
return sum;
}
int main(){
int n ;
printf("enter a value for n :");
scanf("%d",&n);
printf("The sum of first cube of %d natural numbers %d ",n,sum_of_cubes_of_first_n_natural_numbers(n));
printf("\n");
return 0;
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc sum_cub_num.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter a value for n :4
The sum of first cube of 4 natural numbers 100
chetan kumar:~/Desktop/daa programs$
```

Space Complexity:

- The loop doesn't create new memory again and again — it just reuses the same variables.
- Because the memory doesn't grow when n grows, the space stays constant.
- So it is O(1) space.

4) Write a program to find factorial of a natural number using recursive function.

```
#include <stdio.h>
int factorial(int n){
if(n ==0||n==1){
return 1;
 }
else{
return n*factorial(n-1);
  }
}
int main() {
int n;
printf("enter a value for n:");
scanf("%d",&n);
printf("the factorial of %d is %d" ,n,factorial(n));
printf("\n");
return 0;
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc fact.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter a value for n:4
the factorial of 4 is 24
chetan kumar:~/Desktop/daa programs$
```

Space Complexity:

- The function calls itself many times, and each call uses some memory.
- More calls mean more memory is used, so memory increases with n.

- That's why the space complexity is O(n).

5) Write a program to transpose a 3×3 matrix.

```c
#include <stdio.h>
int main() {
int n,m;
printf("enter the size of matrix :");
sScanf("%d %d",&n,&m);
int matrix[n][m];
int transpose[n][m];
int i, j;
printf("Enter elements of the %dx%d matrix:\n",n,m);
 for (i = 0; i < n; i++) {
      for (j = 0; j < m; j++) {
           scanf("%d", &matrix[i][j]);
       }
   }
 for (i = 0; i < n; i++) {
      for (j = 0; j < m; j++) {
            transpose[j][i] = matrix[i][j];
        }
    }
printf("Transposed matrix of given matrix is:\n");
for (i = 0; i < n; i++) {
  for (j = 0; j < m; j++) {
      printf("%d ", transpose[i][j]);
 }
   printf("\n");
}
return 0;
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc trans.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter the size of matrix :3 3
Enter elements of the 3x3 matrix:
2 3 4
4 5 6
3 4 5
Transposed matrix of given matrix is:
2 4 3
3 5 4
4 6 5
chetan kumar:~/Desktop/daa programs$
```

Space Complexity:

- The program creates a 3×3 array, which is fixed in size and does not grow.
- You only use a few extra variables for loops.
- So the space complexity is O(1)

6) Write a program to print the Fibonacci series up to a given number using user-defined function. (For first n natural numbers)

```c
#include<stdio.h>
int fibonacii(int n){
    int x =0;
    int y = 1;
    int next;
     for(int i =1;i<=n;i++){
         printf("%d ",x);
         next = x+y;
         x = y;
         y = next;
     }
     return 0;
}
int main(){
    int n;
    printf("enter a value for n:");
    scanf("%d",&n);
    printf("the fibonacii series is :");
    fibonacii(n);
    printf("\n");SS
}
```

```
chetan kumar:~/Desktop/daa programs$ gcc fib.c -o text
chetan kumar:~/Desktop/daa programs$ ./text
enter a value for n:10
the fibonacii series is :0 1 1 2 3 5 8 13 21 34
chetan kumar:~/Desktop/daa programs$
```

Space Complexity:

- The program only uses a few variables (a, b, sum, temp) and these do not increase when n increases.
- No extra memory grows with the loop.

- So the space complexity is O(1).